Response to the scientific adversarial review

Supplementary response of 21 October 2026

The manuscript now states three claims and a primary use: delivery of an archived H30 magnitude, the value of target-specific diagnostics, the unestablished incremental advantage of window representation, and the separate limits of calibration transport and use. Hot-year classification remains an experimental secondary use. The original H1–H3 definitions and frozen tests are retained; all analyses below are supplementary development on already opened outcomes.

1. **Learning objective and representation.** Script 374 executes the requested quantile/direct-failure-probability × without/with-window experiment with identical reconstructions, information boundaries, capacity, groups and calibration budgets. It includes three seeds, full risk–retention curves, fixed-retention risk and empirical/formal operating points. Script 380 performs another 1,000 complete four-cell learner/calibration/evaluation resamples per protocol. This addresses the omitted matched-capacity classifier without importing old-version outcomes as current evidence.

2. **Use-matched decisions.** Scripts 374 and 381 compare reconstruct all, all unknown, all hot, all cool, matched-retention random rejection, magnitude screening and minimum-cost actions from calibrated hot probabilities. All retain original denominators and report absolute losses, missed/false hot answers and unknowns. Figure 5 now shows absolute loss and deployable alternatives. At FN5/A0.05, original Swiss retrospective losses are 0.4559 for all reconstruction, 0.2775 for expected matched random rejection, 0.1191 for the unchanged magnitude screen, and 0.0500 for all unknown. Source-calibrated direct-feature probability decisions have loss 0.0752 retrospectively and 0.0289 forward. This illustrates scenario dependence rather than a general deployment win. The original weak H3 comparator is not replaced retroactively.

3. **Concentrated failure and information shift.** Script 375 traces every frozen accepted failure by year, gap, sign and available forcing. All 16 CH_2327 failures underestimate H30; 15 use air reconstruction. Low flow varies and is not asserted as their common cause. Official metadata verify station identity and basic catchment attributes, not regulation/groundwater classes. Source donor removal preserves every original mask and regenerates diagnostics; script 377 crosses fitting/calibration information and reports target calibration-group budget curves. Channel loss explains part of the change in reconstruction error, without identifying all international-shift effects or redefining the primary population around a favorable subgroup.

4. **Mechanism denominators.** The original 0.62°C position effect and 0.13°C length effect have different populations and cannot rank their relative importance. Script 376 gives original/common population tables and nested true-window gaps extended left, right or symmetrically with boundary distances. The exact-hidden common subset contains 396 distinct station-years represented by 400 network-target-year rows. Direction-specific nested results show why extension geometry matters, particularly for interpolation. Truth-assisted designs are mechanism assays only.

5. **Trajectories and stronger reconstruction systems.** Script 378 evaluates an explicit slow-residual/seasonal-short-residual candidate using only visible water and available forcing states, reports unsupported cases, common-subset H30 bias/MAE/CRPS, width/coverage, probability scores and 32/128/512-draw stability. Script 379 refits each reconstruction's own failure head, calibration and use decision, distinguishing same-air and flow-enhanced information layers. It does not reinterpret the original fixed eight-draw CSDI reference as a probability-model ceiling. These are exploratory design checks, not evidence that the original window design wins or that all reliable joint models make windows unnecessary.

6. **Deployment populations and a new test.** Script 382 holds target years fixed across 3/5/10-year histories and adds year-start, year-end and fragmented gaps. Script 380 adds target-provider and forward-event holdouts. Artificial masks, calendar replay and real missingness with independent backup truth remain distinct populations. The next-study contract fixes the primary use, a same-information direct-probability comparator, joint risk evidence plus a five-point practical gain, development budgets,

calibration budgets and stopping rules. It does not count 59/118 ideal zero-failure examples as a sample-size plan. Actual new confirmation, independent natural-gap backup truth, verified process types and a real costly acquisition action still require new evidence; none is manufactured from additional masks.

The result-linked tables and figures are generated by script 383, and the follow-up record gives the executed analyses and remaining empirical dependencies. The prospective specification is wrr_next_confirmation_20261021.yaml. A failed or unestablished primary advantage remains part of the final scientific interpretation.

7. **Developed probability budget and a complete system.** Scripts 385 and 390 extend the fixed univariate reference with two air-conditioned CSDI configurations sharing 3,469 old fitting years, thirty old group-disjoint validation groups, five task masks per group, thirty-epoch caps and patience five. H30 CRPS selects the configuration using old validation only. The selected 32-sample H30 distribution is supplied identically to priority, Air2stream-3 and neural points. Native CDF tolerance-exceedance probability precedes the direct/window boosted comparison; every point has its own errors and calibrated admission, and all 945 controlled Swiss test candidates remain. These additions change information and development budget relative to the original reference, and do not establish a probability-model ceiling or reopen a confirmatory claim. Coverage, width, conditional warm/low-flow performance, probability reliability and nested 8/16/32-draw stability are reported.

   The selected distribution covers 81.0% at nominal 90%, with warm-year bias −0.334°C and only 71.0% coverage on the 124 missing-water gaps with complete, majority-low-flow records. On 828 matched point comparisons, MAEs are 0.3455°C for priority, 0.2833°C for Air2stream-3 and 0.3063°C for the neural point. The neural window head accepts 446 cases with 16 failures, compared with 370/6 for the boosted direct head; the native CDF rule already accepts 474/12, giving higher retention and lower sample risk than windows at these operating points. All formal admissions remain empty. At FN5/A0.05, neural direct-head magnitude screening has loss 0.0304, versus 0.0634 for its window head, 0.0530 for the calibrated CDF cost decision and 0.0500 for all unknown. We retain the conditional eight-point increment, but it neither defeats the strongest numerical reference nor produces a general use advantage. These are opened-data comparisons without new independent certification.

8. **Executed joint sample planning.** Script 386 completes 3,000 paired learner refits, recalibration and evaluation under role-specific empirical source laws, over 336 training/calibration/test budget cells. Primary success requires risk evidence for both methods and a Hoeffding lower bound exceeding the five-point gain, with an explicit familywise error allocation. The highest primary success estimate is 3.9%; the normal-approximation sensitivity reaches 13.1%. Simultaneous Monte Carlo intervals support no 80% plan, so we recommend no recruitment sample size from this experiment. Actual fitting/calibration/test supports are 59/21/24 source components; larger simulated budgets create no new histories or climates. This is a conditional planning calculation for the current priority-point heads, not guaranteed power for a new neural design or hydrological population.

   A separate script-405 experiment now covers the developed neural point, the native-CDF reference and the matched direct-boosting reference together. Both score heads are refitted, all three methods recalibrated and paired groups evaluated in 3,000 complete procedures over 432 budget cells. All three risk conditions and both five-point gain lower bounds must hold. Every cell has zero joint successes in 1,000 repetitions; its pointwise 95% Monte Carlo upper bound is 0.37%, and the grid-adjusted upper bound is 0.97%. Table S44 reports this result without treating zero observed simulation hits as proof of zero population power. Actual old role supports are 11/9/7 groups; the probability reconstruction law remains frozen as specified for F2. These old-role empirical simulations provide no new-target recruitment recommendation or independent confirmation. Newly acquired Australian references remain behind the target-error gate.

9. **Public backup search and actual sampling deficits.** Scripts 387–389 stage the public DFO Newfoundland and Labrador archive, freeze metadata and qualification rules before temperature/recovery inspection, and assess 180 candidate sensor-pair-years. None meets the unchanged three-complete-prior-primary-year support. Three compatible-reference pairs failing only historical support represent two primary logger-years, retained in a separately locked short-history sensitivity. All four deficient daily means occur in winter and retain 8–19 hourly measurements; they are not demonstrated whole-day

failures. The hot windows remain observed, explaining zero H30 error in simple references. Air2stream unavailability is reported explicitly. This single dependent provincial cohort of project daily aggregates does not validate risk under long warm-season outages. The user confirms no additional backup or costly auxiliary material is available. Public free backups and air forcing receive no invented acquisition cost, so the query-policy branch remains inactive. Script 391 generates these additional evidence tables without changing the original frozen tests.

10. **Calibration sensitivity and future strong references.** Script 393 holds the developed probability model and development-fitted scores fixed, enumerating all 247 subsets of three/five/seven/nine target calibration groups. Its 741 score/subset evaluations preserve all 945 tests and reproduce the existing full-budget operating points. With three calibration groups the native CDF rule returns no numerical answer in 28 of 84 subsets; nonempty test risk reaches 24.4%, and seven-group subsets still reach 6.2%. These are conditional subset ranges, not confidence intervals, independent observations or joint power. No formal calibration gives positive admission. The next-study contract requires success against both native-CDF and matched-boosting direct references, without choosing a weaker comparator on calibration or test outcomes. Three methods and two paired gain comparisons share the familywise error budget. The additional old-role planning experiment above executes that procedure under its stated finite empirical law; new-target confirmation remains unexecuted. Neither the original H1–H3 nor the interpretation of the original paired power experiment is rewritten.

11. **Additional reference intake and dependence checks.** The new Australian pool provides 80 station-years at 52 stations passing the fixed daily-quality and native-sampling requirements; target prediction errors remain unopened. Original polygon-based role counts of 20/16/10 are qualification metadata, not verified independent deployment units. A complete cached-network weak-connectivity screen accounts conservatively for shared upstream bifurcations and ungauged confluences: 516 mapped candidates form 147 hydrologic blocks. Mapped qualified references occupy 33 blocks, six spanning original roles, while one qualified station remains pending topology review. Within-role counts of 16/15/9 can share a block across roles and must not be added. Roles and reference qualification files are preserved. The mapped pool alone cannot reach even the ideal 208 disjoint calibration-plus-evaluation-group floor under the future error allocation; 984 unmapped candidates remain unresolved. These necessary floors are not recruitment recommendations. The added SI subsection reports this scope without promoting acquisition into independent confirmation or concluding that conservative basin connectivity proves record leakage. All 11 conservation checks pass; weather-event dependence and prior-pool exclusions still require verification.

12. **Conditional-law and complete-system attribution.** Tables S45–S50 separate conditional-law quality from the delivered point, score learning and use decisions. Seven laws and three heads give 21 comparisons at one unchanged priority point; the complete system adds 24 same-air and 18 separate flow-tier cells, refitting each numerical point's own error. On 907 common controlled Swiss cases, the development-selected seasonal/air residual model has CRPS 0.210°C versus 0.273°C for raw blocks. At 32 draws its nominal 90% band covers 85.7%; at 512 draws aggregate coverage approaches 90%, while warm-anomaly coverage remains 84.6% and median bias −0.379°C. All 63 formal calibrations remain empty. These diagnostics on previously opened data preserve the original hypotheses, prospective predictor and sealed USGS/Australian error gate. Full-pool costs retain all 945 candidates and include direct cost, constant and random references.

13. **Additional reference feasibility, without opening new errors.** The first USGS metadata plan has attempted all 1,615 jobs; 116 responses still lack a usable verified reach index or matching coordinates, and 155 retained candidates are not qualified independent references. A separate fixed extension of 157 prior identities verifies 129 additional exact-identity/location/reach records while retaining unindexed or unresolved cases; the retained candidate count remains 155. The current retained USGS and quality-qualified Australian pools have an optimistic upper bound of 189 units, below the necessary ideal disjoint calibration/test floor of 208 under the fixed error allocation, even before training and actual power. This is a roster-specific necessary screen, not a recruitment recommendation. DFO coverage remains at most two complete prior years even under permissive cross-sensor hourly unions. Further no-login NorWeST and EcoDrought metadata/access checks have not released an eligible backup: 29

published NorWeST daily links returned HTML headers after redirection; EcoDrought metadata span September 2018–September 2021 and its archive alone cannot supply three complete prior years. HTTP failures and unresolved custody are not counted as zero observations. These source reports preserve all original eligibility and prediction locks.

Original response of 20 October 2026

This responds to the internal scientific assessment supplied for this revision; it is not an official WRR editorial decision. The previous manuscript and response are preserved in the 20261020b archive. The new title is **When are incomplete river-temperature records fit for estimating thermal extremes?**

The revision executes the proposed experiments rather than treating additional code or successful builds as scientific answers. It does **not** establish the proposed uncertain-window representation's practical superiority. Unchanged transfer also exceeds its empirical risk target. The manuscript is rewritten around these findings rather than retaining the former method-success narrative.

The protocol, hypothesis table, five-point superiority and two-point equivalence margins, source/target roles, cost grid and stopping decision are in the revision documents. The method lock precedes target calibration errors; the deployment lock precedes new-test errors. A later descriptive snow-unit correction is explicitly recorded and changes no fitted prediction or threshold.

1. Prior work and the actual contribution

We withdraw any implication that previous thermal-data studies considered missing-day counts alone. Johnson et al. (2021) considered timing, and Beaufort et al. (2022) directly reconstructed thermal peaks with seasonal information. Selective prediction with a specified risk target and learn-then-test calibration are also prior statistical ideas. They are credited in the new Introduction and Methods.

The contribution is a fit-for-use assessment connecting paired thermal recovery, an accurately interpreted maximum-statistic identity, matched-information diagnostics, frozen transfer and original-candidate decisions. It is not a new general selective-learning algorithm. A proposed uncertain-window representation is a tested hypothesis, not a demonstrated superior design.

2. Explicit overlap, implicit target information and stronger alternatives

The revision compares H30 disagreement with daily disagreement from the same reconstruction information, and adds seasonal splines, actual reconstruction OOF residual diagnostics, historical mask-matched H30 errors and direct H30 uncertainty. It includes both the current-policy-aligned certificate and a whole-system certificate that selects and calibrates its own delivered point.

On documented-mean source admission, uncertain-window gain over the generic H30 score is 1.7–3.8 percentage points. Its conditional difference from direct H30 uncertainty lies within the predefined two-point band, but whole-procedure uncertainty does not establish equivalence. The complete diagnostic model is competitive too. We do not translate a confidence interval crossing zero into equivalence or interpret a feature deletion as removing implicit peak information.

There is a useful positive result: H30 rather than daily disagreement adds 3.8–6.2 source points and 9.4 points [6.1, 13.1] on independent Swiss records under local recalibration. Both risk point estimates in that latter comparison are below 5%, but their uncertainty does not certify the cap. Explicit overlap improves a restricted logistic model, not consistently a spline. This supports target-specific diagnosis, not the necessity of a particular peak-window representation.

3. Representativeness and the experimental population

The mechanism experiment no longer selects the first twenty-four stations of two providers. It uses all eligible opened records: 406,015 retrospective masks and 290,322 earlier-year masks across 193 labels, 581 provider-

station identities and 6,170 distinct station-years. All target and donor identities reduce those labels to 181 overlap components. These components are not asserted to have independent climates.

The primary documented-mean controlled analysis has 312,825 masks on 4,720 station-years in 144 components. Repeated starts, within-warm controls, equal-budget fragmentation and infeasible-cell accounting are explicit. At exact true-peak hiding, the documented-mean length contrast uses 396 station-years in 98 components rather than the earlier eleven station-years.

We also corrected a design-level mistake before final testing: a 30/60/90-day mechanism panel is not an appropriate primary admission population. The new admission assay includes short gaps and empirical shapes: 70,011 retrospective and 46,510 earlier-year cases. Earlier controlled-panel classifier gains are retained as diagnostic history, not the headline. The mixture remains an assay rather than an estimate of natural monitoring outage prevalence.

4. Window switching and the sign of the error

The term previously called a false-peak channel is now the nonnegative window-switch increment. The manuscript gives its exact identity and distinguishes net overestimation from compensation for underestimation. In the primary climatology analysis, 60.4% of positive increments still end below true H30. A coarse aligned/misaligned contrast is not identified with this term.

The claim that H30 is damaged only when its true window is hidden is removed. With fully observed true support, another reconstructed window can still exceed it. Length is not dismissed: extending a fully hidden true peak from a 30- to 90-day outage adds 0.13°C [0.10, 0.17] for climatology and 1.24°C [1.12, 1.35] for interpolation.

5. Oracle diagnosis versus deployable improvement

Residual permutation and mean removal are crossed as a two-by-two intervention. Primary climatology MAEs are 0.450°C originally, 0.398 after permutation, 0.215 after mean removal and 0.071 after both; the interaction is −0.091°C. Their percentages are not additive physical-mechanism contributions.

The separate visible-information correction improves policy MAE by only 0.014°C [0.008, 0.019]. It does not demonstrate the large operational improvement suggested by oracle centering. Conditioning on the true maximum can itself induce negative error; general physical cold bias is not inferred from that selected subset.

6. Risk semantics, zero failures and concentrated errors

The empirical bootstrap selector is explicitly heuristic. Four thousand constructed stress repetitions include nearly perfectly correlated within-group failures, uninformative scores, zero-failure calibration and a rare bad group. In the twelve-group population with true failure probability 6%, the empirical selector violates the 5% target in 45.4% of repetitions. This is not an estimate of river failure frequency.

A separate bounded-group KL test targets the ratio of expected group failure-and-admission to expected group admission. It allows arbitrary within-group dependence but requires independent deployment-representative groups, an independently fitted score and training-fixed candidates. It neither protects every subgroup nor survives arbitrary shift. Even zero failures with full admission need 59 groups for one threshold and 118 for twenty-one. Available calibration cannot certify nonzero admission. Empty admission is not risk zero.

Pooled, equal-component, station-year-weighted and per-component outcomes are retained, including denominators and unavailable estimates. No binomial case interval is claimed for repeated masks in a single dependent river.

7. Frozen transfer and genuinely prospective evaluation

Continuous reference histories were reacquired from original agencies after metadata-only splitting. The 177-catchment extreme resource was not mistaken for complete daily truth. The all-old-target/donor audit and

strict completeness leave 53 stations: 21 development, 13 calibration and 19 test, including seven Swiss and twelve French test components. French aggregates remain a distinct sensitivity scope.

F0 transfers both head and source threshold; F1 changes only local calibration; F2 fits on new development records and calibrates locally. Target-provider exclusion and USGS-only intercontinental source restrictions are separate checks. Local historical reconstruction is allowed as stated, not selected by hidden-year error.

Primary Swiss F0 keeps 56.1% of 669 cases but has 8.8% selected failures. F1 keeps 47.5% with 4.1% point risk, and F2 44.4% with 3.0%. The F1 risk interval still exceeds 5%. The calendar-time forward head/calibration use years through 2018 and every local training input is earlier-year-only for 2019–2022 evaluation. Inputs are current-vintage observations/reanalysis, not archived real-time publication vintages. Its Swiss F0 point risk is 9.8%. Small-group intervals do not prove population risk is above 5%, but do not validate unchanged risk transport either.

8. Independent calendar replay and the no-screen claim

The composition bins were inherited from opened analyses, but rates are fitted on opened-source empirical geometries and frozen before new-station calendar replay. The final paper does not claim that the earlier 67 target station-years were independent of development or that absence of target-calendar labels made that analysis blind.

New eligible natural calendars are French aggregates: 168 masks in ten components, with 21.4% unscreened failure against a composition prediction of 26.6%. Their conditional population differs from the old replay; this does not estimate failure across all monitoring years, many of which have no gaps. No qualifying Swiss documented-mean natural-calendar estimate is available. Most-networks-no-screen advice is withdrawn; a network point forecast cannot certify its risk.

9. Hot/cool refusal and the downstream denominator

We evaluate local training-q90 hot labels, fixed 18/20°C labels, threshold proximity, false-hot/missed-hot outcomes, hot/cool refusal, unknowns, coverage and selected-sample frequency distortion. No unknown record is labelled cool, deleted from the denominator or assumed perfectly repaired.

For primary Swiss F0, hot, cool and overall refusal are 46.4%, 43.4% and 43.9%, so the old no-disproportionate-hot-refusal conclusion is not universal. At declared missed-hot cost five and unknown cost 0.05, loss changes by -0.3368 [-0.4059, -0.2720] per original candidate. Other costs can reverse the benefit. These are cost scenarios, not measured ecological benefits or simultaneous claims across the sensitivity grid.

A full observed-window lower bound can identify a hot label without accepting H30 magnitude. The branch is valid relative to the recorded reference, not a certified sensor-error model. It rescues only two primary retrospective and one prospective labels. It is not presented as a large new design advantage.

10. Missing final-protocol work and evidence closure

A post-test deterministic baseline audit also compares always-unknown/hot/cool answers under the unchanged cost grid. At FN5/A0.05, the primary screen's absolute loss is 0.1191, higher than always unknown at 0.05. A large benefit over reconstructing all does not establish operational desirability; some other declared cells beat both at their point estimates, not as a new confirmatory endpoint. No predictor, threshold or original H3 comparator changes.

The formerly missing work has been executed: broader controls; 324 source score tasks; conditional trajectories and direct-H30 alternatives; actual-reconstructor diagnostics; event/cofailure and natural-calendar assays; continuous new histories, discharge and published-event matching; F0/F1/F2 and provider/intercontinental transfer; Air2stream and CSDI references; repeated learning curves; and 1,000 complete learner/calibration resamples per protocol. Multiple masks and split seeds are never counted as new hydrological replications.

The primary five-point representation hypothesis is not supported. Whole-procedure gain intervals include zero and more target groups alone do not solve training/calibration uncertainty. We stop tuning rather than

changing the comparator, margin, role assignment or headline population to obtain a win.

A snow-percentage label error was corrected descriptively after the first frozen report. Neither features nor decisions use it; the immutable locks and original outputs remain unchanged, and corrected hydrological tables supersede the raw labels. Groundwater and regulation classes are unverified, not invented.

The manuscript, SI, five main figures, two supplements and generated tables now describe these final experiments. Source bindings and scientific completion checks are separate from software tests. Author details and a permanent analysis archive still require verified human input; scientific revision is not a claim of submission readiness or guaranteed WRR acceptance.